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A Residual Decomposition Framework with Attention-Based Sequence Modeling for Renewable Energy Forecasting and Blockchain-Anchored Scheduling

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ABSTRACT The need to accurately predict renewable energy production over a short time period is important in order to operate a smart grid. It also allows for better coordination of energy storage and provides an opportunity to improve how efficiently energy is produced and used through "energy aware" scheduling. Forecasting errors are typically larger during the periods when there is higher variability in renewable generation output. Therefore, this research introduces a residual decomposition based predictive model that integrates an easily understood linear base line model with a low computational cost attention-based Bidirectional Long Short Term Memory (BiLSTM) residual learner. A blockchain-based oracle layer has been included in the design of the model so that forecasts can be utilized as anchor points for scheduling. The basis for designing the above described model was the results of analyzing failures of using Recurrent Neural Network (RNN) models independently as well as the numerous architectures developed by attempting to refine these independent RNN models. As previously stated, the primary focus of this study is to develop a residual decomposition-based forecasting model. In developing the model, the forecasting problem was decomposed into two component parts. The first part is a deterministic and linear component and was modeled using Multivariate Linear Regression (MLR). The MLR model represents the majority of the structure found in the renewable generation data collected for Tamil Nadu and includes both the major linear trends and diurnal patterns. The second component part of the problem is a stochastic component and is represented by a BiLSTM model. The BiLSTM model will learn from only the residual error left after applying the MLR model to all variables. Data for testing were created using eight thousand seven hundred sixty hours of actual renewable generation data collected from solar panels located at four sites in India. Real world generated loads and storages have been added to these data sets. These data sets provide realistic physical groundings for each of the three variables (load, renewable generation, and energy storage) that are being modeled. The proposed model achieved a Root Mean Square Error (RMSE) of .007337 and Mean Absolute Error (MAE) of .005312, while achieving an R-Squared value of .98805. When comparing the performance of the proposed model to that of Random Forest, it can be concluded that the proposed model performed approximately twenty-five percent better than Random Forest in terms of RMSE and forty-four percent better than Random Forest in terms of unexplained variance. Finally, the oracle layer within the proposed model serializes predictions into canonical JSON objects. Each object is then hashed through Inter Planetary File System (IPFS), blockchain hashing, and then exposed to smart contracts as verifiable control signals. System-level evaluation shows that forecast-gated operation increases renewable utilization from 15.34

INDEX TERMS Attention mechanism, bidirectional LSTM, blockchain oracle, energy-aware scheduling, residual decomposition, renewable energy forecasting, smart grid.

I. INTRODUCTION

RENEWABLE energy forecasting is now a core requirement for power-system operation because solar and wind production varies with weather, time of day, and season. As renewable penetration increases, grid operators must make dispatch, storage, and scheduling decisions under greater uncertainty. Forecast error during peak-generation or transition periods can lead to underutilized renewable energy, unnecessary curtailment, increased reserve requirements, and inefficient downstream decisions [1]–[6]. These effects are especially important in smart-grid settings where energy-aware compute scheduling, peer-to-peer energy exchange, and distributed control depend on forecast signals that are both accurate and auditable.

As such, this research developed a new approach to forecasting the total renewable energy output in a given area. To achieve this goal, the research team used a residual decomposition methodology. As opposed to a traditional approach where all variables are fed into a singular predictive algorithm, the residual decomposition methodology splits the forecasting process into two components. First, the model predicts the major renewable energy generating trends using a straightforward and easy-to-understand statistical model, in this case, Linear Regression. Second, the residuals generated by the initial model are then fed into another model that focuses solely on predicting any remaining non-linear patterns in the data. The second stage of this methodology employs a Recurrent Neural Network (Bi-LSTM) to learn from historical renewable energy output data and predict future values. The RNN was designed to look at the data over short-term horizons (typically minutes to hours). Since there is no need to include past values that have been explained by the first model, the RNN only needs to analyze small sequences of hourly values (typically around 10-15 minutes). By limiting its scope to analyzing relatively short sequences of hourly values, the RNN becomes much smaller and easier to train compared to other types of neural networks. Additionally, the use of attention weights allows the network to prioritize which hourly values are most relevant for making forecasts.

Lastly, after each hour's value has been predicted, the model produces a green score. The green score represents how confident the model is in its predictions. The scores range from 0 to 100, and higher scores represent greater confidence. The green score serves as a threshold value for smart contracts. If the green score exceeds a predetermined threshold level (e.g., 80), the contract is executed. Otherwise, it is rejected. This prevents unnecessary trades from occurring based on poorly supported predictions. To further support the development of this platform and ensure trustworthiness within multi-party environments, the authors integrated their forecasting model with an oracle-style blockchain layer. The oracle layer records forecast payloads in canonical JSON format and anchors their digital fingerprints onto the blockchain. The validated forecasts are then available as inputs for renewable-aware scheduling and trading decisions.

Therefore, the proposed method includes: * ML Forecast → Canonical JSON → IPFS CID → Hash Anchor → Blockchain Validation → Trading/Scheduling. * Blockchain is not just a separate tool applied after prediction but instead serves as the underlying decision-making infrastructure allowing various stakeholders to utilize tamper-proof forecast signals for scheduling/trading/execution validation.

Overall, this research contributes to three key areas:

1) Introduces a residual decomposition framework for renewable energy forecasting separating deterministic structure from stochastic deviations enhancing model interpretability and learning efficiency.
2) Proposes a lightweight attention-based BiLSTM architecture operating on residual error to enable focused learning of nonlinearities and peak-hour anomalies.
3) Integrates forecasting with blockchain-based oracle mechanisms to create tamper-evident anchorages of predictions serving as verifiable control signals in decentralized renewable-energy trading/scheduling systems.

Through empirical evaluations utilizing real-world data collected from the Tamil Nadu Grid Corporation Ltd, this research demonstrated that scheduling renewables based upon accurate forecasts can lead to significant increases in overall renewable utilization.

Additionally, this research demonstrated empirically that residual-learning achieves similar levels of performance as strong linear-baseline approaches while offering better performance in nonlinear operating regimes.

II. LITERATURE REVIEW

A. RENEWABLE ENERGY FORECASTING

Renewable energy forecasting has been studied using statistical, physical, machine-learning, and deep-learning approaches [3], [7]–[12]. Classical time-series methods such as ARIMA and SARIMA provide interpretable temporal baselines and remain useful when seasonality is stable [13], [14]. However, renewable generation is affected by nonlinear interactions between irradiance, wind speed, installed capacity, and operational constraints. Machine-learning methods therefore became popular because they can map nonlinear feature interactions without requiring a full physical model [18]–[20].

B. CLASSICAL MACHINE-LEARNING MODELS

Linear Regression is often competitive in structured energy datasets because generation profiles contain strong deterministic patterns from diurnal cycles and installed-capacity effects. Random Forest, extremely randomized trees, and gradient-boosted methods can capture nonlinear relationships and feature interactions [21]–[25], but they do not directly model temporal sequence order unless lagged features are manually encoded. In the present work, both Linear Regression and Random Forest are used as baselines, while the proposed model uses the linear component as a deterministic anchor rather than treating it only as a competitor.

C. RECURRENT SEQUENCE MODELS

The long short-term memory (lstm) network was developed to deal with the issue of vanishing gradients in recurrent models [26]. Additionally, gated recurrent units, encoder-decoder SEQUENCE MODELS, and probabilistic recurrent forecasters extended the application of deep sequence models for predicting time series based on sequences [27]–[29]. Furthermore, bidirectional recurrent networks expanded upon the previous idea of using a SEQUENCE model to process a sequence in both forward and backward directions to create richer contextual representations over the input window [30]. In forecasting, bilstm MODELS can utilize the temporal context from windows of historical measurements. However, a standalone recurrent learner could be inefficient when the target includes a large deterministic portion. Therefore, the recurrent model would need to learn both the smooth diurnal pattern and the irregular weather driven deviation simultaneously. Learning this information simultaneously can result in less efficient parameters, less effective convergence rates, and less stable improvements compared to other simpler baselines. Residual learning is able to overcome the issues created by inefficiencies resulting from learning both patterns simultaneously, because it requires the SEQUENCE model to focus solely on learning the deviations about a strong baseline [31]–[33].

D. ATTENTION AND TRANSFORMER-BASED MODELS

Models utilizing Attention mechanisms are capable of weighting different temporal positions in relation to how relevant they are to the task being modeled [34], [35]. The transformer architecture expands upon this concept via self-Attention and has resulted in many successful applications within sequence modeling [36]. Variants of the temporal series transformer architecture, including temporal fusion transformer, informer, autoformer, FED-former, PatchTST, and TimesNet have also improved upon multi-horizon forecasting and long-sequence modeling [37]–[42]. However, traditional transformer architectures require quadratic memory growth with increasing sequence lengths and generally require larger training sets than standard models. They also make significant sacrifices regarding compactness that makes edge deployment appealing. Given moderately sized datasets and real-time deployment scenarios, lightweight Attention-based recurrent models may serve as a more practical approach to achieve comparable results to full transformer architectures. Preliminary experiments conducted on the Tamil Nadu task demonstrated that multiple energy aware Attention variants produced minor marginal improvement before reaching a plateau – Attention added additional optimization complexity and increased the risk of overfitting instead of providing net performance gains. Due to these findings, the decision was made to pursue a residual decomposition strategy instead of continuing to develop further standalone Attention.

E. BLOCKCHAIN-BASED ENERGY SYSTEMS

The use of blockchain technology has also received a significant amount of attention in terms of Transactive Energy (TE), Peer-to-Peer (P2P) Markets, Smart Contracts, and Decentralized Settlement [43]–[51], all of which enhance system-wide transparency by removing the dependence upon a single trusted entity. Most however rely upon historical meter reads or ex-post transactional logs. In contrast prior energy trading platforms have primarily focused on Bidding, Clearing, Escrow and/or Settlement; while predictive accuracy is typically the focus of forecasting studies. Only a few research efforts provide both an auditable link from predictive payloads to their associated anchoring within Interplanetary File System (IPFS) and enforcement via On-Chain Gating Logic. Therefore Trustworthy Oracle Mechanisms, Decentralized Data Feeds and Authenticated Off-Chain Computation are important considerations with regards to Blockchains that rely on Physical World Measurements or Machine-Learning Predictions [52]–[56]. As such there is a need for a Secure Auditable Framework whereby Predictive Models may function as Trusted Inputs to Decentralized Decision-Making Processes. The above gaps are summarized in Table 1.

F. REASEARCH GAPS AND PROBLEM FORMULATION

Three coupled gaps exist among the reviewed literature that motivate this study. First, although prior renewable forecasting studies have treated the full generation signal as a singular target for learning capabilities of nonlinear MODELS, this forces those same MODELS to re-learn deterministic diurnal structures that simpler baselines already effectively capture efficiently; this waste of model capacity reduces robustness in high variability environments. Second, there is little documentation of failure modes among prior evaluations of deep learning for energy forecasting. That is to say, when standalone recurrent MODELS perform worse than linear baselines, there is rarely documented evidence on why such occurred or what architectural modifications were attempted; thus practitioners lack actionable knowledge. Finally, blockchain-based energy platforms and forecasting research are largely independent of one another; energy trading systems document transactions but do not document verifiable forecasts, whereas forecasting studies document accuracy metrics but do not provide an auditable means of enforcing signals for scheduling and trading purposes. There is no currently available integrated framework that serializes ML forecast payload outputs into distributed storage systems and utilizes them as on-chain gate logic for determining trading decisions. The proposed residual decomposition framework with blockchain oracle integration provides solutions for each of these three gaps.

III. SYSTEM OVERVIEW

A. INTEGRATED FORECASTING AND DECISION ARCHITECTURE

As shown in Fig. 1, the proposed integrated framework links forecasting, validation and scheduling within one decision

TABLE 1. Comparison of Existing Models and Research Gaps

Model / Approach	Key Strength	Limitation	Gap Identified	Proposed Advantage
Linear Regression (LR)	Simple, interpretable, and strong for structured deterministic data	Cannot capture nonlinear and peak variations	Poor behavior in critical nonlinear regions	Residual learning models nonlinear deviations separately
Random Forest (RF)	Captures nonlinear relationships and feature interactions	Lacks explicit temporal sequence modeling	Ignores ordered dependencies in energy data	BiLSTM captures temporal dynamics effectively
LSTM	Models temporal dependencies through gated recurrence	Learns the full signal and may spend capacity on deterministic structure	Overburdened by deterministic and stochastic components	Residual decomposition reduces learning complexity
BiLSTM	Captures bidirectional temporal context	Still processes the full target uniformly when used alone	Limited focus on high-impact deviations	Attention emphasizes influential residual windows
Transformer	Powerful self-attention and parallel processing	Higher computational cost and data demand	Less suitable for moderate datasets and edge use	Lightweight attention-BiLSTM provides an efficient alternative
Blockchain Energy Systems	Transparency, decentralization, and auditable settlement	Often depend on historical or static data	Forecast signals are not integrated as verifiable control inputs	Oracle-based anchoring enables trusted forecast-driven decisions

pathway. Renewable data from Tamil Nadu is used to create both cyclical and physical features from modeled and observed operating data. The residual structure learned by the attention based BiLSTM model plus a linear baseline represents the dominant deterministic relationship. The corrected forecast along with some additional information (metadata) is then forwarded to an oracle layer for validation/anchoring prior to being consumed by either scheduling/trading logic. Thus, the key architectural contribution is that these three layers (forecasting, blockchain decision, and actuation) were developed together. Forecasting was not treated as a secondary product; rather, it is the primary operational variable determining if trading or energy intensive execution should occur.

B. BLOCKCHAIN AS DECISION-LAYER INFRASTRUCTURE

In the design of this integrated framework, we have chosen to utilize blockchain as our decision-layer infrastructure due to its ability to provide trust, transparency, and tamper-proof validation in multi-party energy systems. With respect to traditional central-architecture models, which require a single, trusted authority to disseminate forecasts and make control decisions, blockchain enables decentralized verification of forecast integrity. Through the process of hashing (via cryptographic hashing) and distributed-consensus mechanisms, this system provides assurance that forecast-driven scheduling and/or trading decisions are auditable and cannot be manipulated by any actor. This is particularly important in peer-to-peer energy networks, renewable-aware compute-scheduling environments and other decentralized demand response applications where numerous actors will need to operate under a set of shared forecast signals without requiring a single entity responsible for creating all of those forecasts.

The mechanism of integrating the two is as follows: after forecasting has occurred, a validation node bundles the forecasted value along with the current snapshot of relevant feature attributes, timestamp and the metadata associated with the current schedule eligibility into a canonical JSON object. It is crucial to serialize this object deterministically because standard JSON objects may vary by key ordering, whitespace, etc., depending upon environment; failure to do so could result in identical logical forecasts producing different hash values. Once serialized, the payload is written out-of-chain to the Interplanetary File System (IPFS), which returns a Content Identifier (CID). At the same time, the payload is hashed using a compatible Ethereum Virtual Machine (EVM) function (keccak256) and only the compact fingerprint of the payload along with the CID and associated renewable score are stored on-chain to prevent blockchain bloat while maintaining full verifiability.

Therefore, the oracle layer performs four functions. First, it validates the forecast payload and metadata. Second, it calculates a cryptographic digest of the forecast object. Third, it anchors the digest to a ledger record via consensus. Finally, it exposes the validated forecast as an input to decision-layer scheduling and/or trading logic. There are three Smart Contracts making up the on-chain components of the platform: PredictionStorage anchors the IPFS CID and keccak256 hash for each forecast payload providing immutable provenance; EnergyTrading utilizes the forecast gated Predictive Green Score to govern or restrict order execution and escrow to only allow trades when the PGS is greater than the on-chain confidence threshold of 7000 on a 10000 integer scale (solidity compatible fixed point representation); finally there is an overlay TradeDispute Smart Contract managing disputed settlements through a win/lose resolution mechanism. The platform utilizes an ENRG native token to denote energy trade settlement amounts thereby linking financial incentives with renewable availability signals. Therefore, the entire pipeline is: ML Forecast -> Canonical JSON -> IPFS CID -> Hash Anchor -> Blockchain Validation -> Trading/Scheduling.

IV. METHODOLOGY

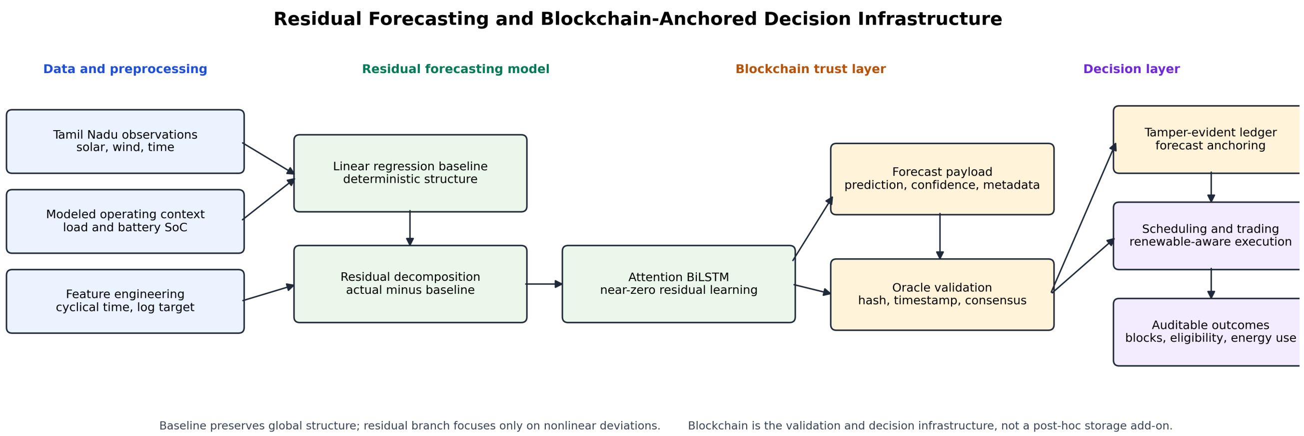

FIGURE 1. **Residual forecasting and blockchain-anchored decision infrastructure. The figure shows the full path from Tamil Nadu renewable observations to residual forecasting, oracle validation, ledger anchoring, and scheduling. The key insight is that blockchain acts as the trust and decision infrastructure for forecast use, not a detached storage layer. The conclusion is that forecasts become auditable control signals for decentralized energy operation.**

A. PROBLEM FORMULATION

Let $\mathbf{x}_t$ denote the multivariate feature vector at time t, including renewable capacity factors, generated power, modeled load, modeled state of charge, and cyclical time encodings. The target y_t is the normalized renewable-generation signal in log-transformed form:

$$y_t = \log\left(1 + \frac{P_t^{gen}}{P^{peak}}\right), \tag{1}$$

where P_t^{gen} is total renewable generation and P^{peak} is the peak-load normalization constant. In addition to using $X_{t-L+1:t}$ as an input to estimate y_t, the model also uses the data from $X_{t-L+1:t}$ with a look back window L. The main issue facing this problem is that the series can be viewed as having both linear and nonlinear components. A majority of the component that represents the state in the series is a function of time and includes a strong periodic nature due to its association with the sun's position relative to earth and to the periodic nature of wind patterns. This portion of the series will have a very high degree of predictability. However, there are some other portions of the series that reflect abrupt changes (due to unusual localized weather conditions) or variations in load on the storage systems and these represent a much lower frequency of occurrence but are operationally significant. If the model does not account for the predictable aspects of the series it would essentially be "re-learning" obvious patterns. On the other hand if the model ignores the less predictable aspects of the series it would produce systematic errors during peak usage periods when energy decisions are most important.

B. DATASET CONSTRUCTION

The core renewable generation data used in this study is obtained from real-world hourly observations representing Tamil Nadu grid conditions. To evaluate system-level interactions such as scheduling and storage dynamics, additional variables including load demand and battery state are incorporated using physically grounded modeling approaches.

The raw renewable data are drawn from the GridPath India long-term power-system planning dataset, which provides hourly capacity-factor traces for solar and wind resources [57]. The underlying resource traces are consistent with common renewable resource data products and reanalysis workflows used in wind and solar studies [58]–[61]. Tamil Nadu records are filtered from fixed-tilt solar PV, single-axis solar PV, rooftop solar PV, and adjusted existing wind folders. Each source is checked for hourly continuity, missing timestamps, duplicate timestamps, and null values. The final hourly renewable table contains 8,760 observations for calendar year 2019.

Modeled load is represented with a deterministic daily profile bounded between 7,000 MW and 14,000 MW. Modeled battery state of charge (SoC) follows configured storage limits, charge/discharge efficiencies, and maximum rate constraints. These modeled components are introduced to provide a consistent evaluation environment for the proposed forecasting and scheduling framework and do not replace the underlying real-world generation data.

C. PREPROCESSING PIPELINE

The preprocessing pipeline performs deterministic file discovery, schema validation, Tamil Nadu filtering, timestamp parsing, per-folder aggregation, and final merge. Capacity factors are averaged across Tamil Nadu projects at each timestamp. Cyclical features are encoded as

$$h_t^{sin} = \sin\left(\frac{2\pi h_t}{24}\right), \quad h_t^{cos} = \cos\left(\frac{2\pi h_t}{24}\right), \tag{2}$$

TABLE 2. Dataset and Feature Summary

Item	Description
Temporal coverage	8,760 hourly observations from January 1 to December 31, 2019
Region	Tamil Nadu, India
Renewable inputs	Fixed-tilt solar, single-axis solar, rooftop solar, and adjusted existing wind capacity factors
Modeled variables	Load demand, battery SoC, and scheduling eligibility variables
Target	Log-transformed renewable generation, $\log(1 + P^{gen}/P^{peak})$
Feature set	Renewable factors, P^{gen}, P^{load}, SoC, hour sine/cosine, and day sine/cosine
System-level data	A 15-minute, five-node engineered trace in `data_processed` for oracle-facing prediction streams

with analogous encodings for day of year. These encodings prevent artificial discontinuity between adjacent hours such as 23:00 and 00:00.

D. RESIDUAL DECOMPOSITION LOGIC

The residual decomposition first trains a Linear Regression model f_{LR}:

$$\hat{y}_t^{LR} = f_{LR}(\mathbf{X}_{t-L+1:t}). \tag{3}$$

The residual target is then defined as

$$r_t = y_t - \hat{y}_t^{LR}. \tag{4}$$

While this approach transforms the forecasting problem into a residual model for residuals, the large part of the overall deterministic nature is modeled using the linear model as well. Thus, the sequence model can focus solely on the stochastic errors in the data stream. By removing much of the variation and complexity from the original target signal through its reduction to a residual, the learning process becomes more efficient and stable regarding convergence of the non-linear models. By decomposing the learning problems in such a manner, we can ensure that the non-linear model will learn to recognize local anomalies based upon their proximity to zero (near-zero mean) rather than attempting to reproduce the entire signal. As confirmed during the notebook run, when trained, the mean of the training residuals is approximately 0.000000 indicating that the residual models operate at or very near a target value of zero. The concept is similar to decomposition-based forecasting as well as residual learning where different types of trend, seasonality, etc. are separated to minimize complexity of targets [15]–[17], [31], [32]. Statistically speaking, this will reduce the variance of your targets. Your original targets contain some form of smooth deterministic structure as well as sparse anomalous behavior. Isolating these anomalies within a residual provides you with a lower variance learning problem which is significantly easier to optimize. From exploratory analysis of residuals in regards to testing identified 64 extremely non-linear points out of 1266 sample points taken from testing. That represents 5.1 percent of all sample points. There was no uniform distribution of these points; they were concentrated at high-generation periods and time transitions. At these times, the operational cost of a miscalculation would be greatest. Therefore, while there was not an imposition of a two stage architecture as previously described, we found it to be evidenced — Linear Regression provided sufficient explanation for modeling the base trends and our residual error analysis showed us that the remaining problem was smaller and more localized and thus better suited for targeted sequence learning.

E. ATTENTION-BASED BILSTM RESIDUAL LEARNER

The residual learner receives a 48-hour lookback sequence with 10 input channels. A bidirectional LSTM layer extracts temporal features from both forward and backward directions. The bidirectional structure is motivated by the observation that renewable generation is influenced not only by the most recent hours but also by where the current hour lies within the surrounding diurnal pattern: a bidirectional encoder can inspect both the local lead-in and the lagged tail of the window during representation building. The attention scoring function first computes a saliency score for each hidden state:

$$e_i = \tanh(\mathbf{W}_a \mathbf{h}_i + \mathbf{b}_a), \tag{5}$$

which is then normalized via softmax to produce attention weights and an attention-weighted context vector:

$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^{L} \exp(e_j)}, \quad \mathbf{c} = \sum_{i=1}^{L} \alpha_i \mathbf{h}_i, \tag{6}$$

where $\mathbf{h}_i$ is the hidden state and $\mathbf{c}$ is the attention-weighted context. The context vector is projected to the residual estimate through a dense projection:

$$\hat{\varepsilon}_t = \mathbf{W}_o \cdot \varphi(\mathbf{W}_c \mathbf{c}_t + \mathbf{b}_c) + \mathbf{b}_o, \tag{7}$$

where φ denotes ReLU activation. Attention is included because not every hour in the lookback window contributes equally to the residual: peak-generation errors are often driven by the interaction between immediate short-horizon fluctuations and periodic diurnal recurrence. The mean attention profile observed in development concentrated most strongly on the most recent one to two hours while also assigning informative weight near the daily recurrence boundary (approximately 24 hours prior). This profile (the network that emphasizes the most current data with the least dependency on approximate daily recurrence) illustrates the physical nature of the residual learning issue; sudden changes in the weather occur within the last few hours while the structural influence of predictable daily solar influences remains. There are three advantages to this additive form: the terms have an interpretable function, therefore they preserve interpretability; it makes optimization easier as it converts the deep part of the network from predicting targets to correcting errors; and it prevents degradation since if the residual component

TABLE 3. Attention-Based BiLSTM Residual Learner Architecture

Layer	Output Shape	Parameters
Input	(48, 10)	0
Bidirectional LSTM	(48, 32)	3,456
Attention layer	(32)	80
Dropout	(32)	0
Dense	(16)	528
Output dense	(1)	17
Total	–	**4,081**

predicts nearly nothing, then the model will revert back to its previously validated linear base. The results are transformed inversely to their original scale to be evaluated.

F. MODEL ARCHITECTURE

Table 3 reports the exact architecture obtained from the notebook. The model has 4,081 trainable parameters, making it substantially lighter than typical Transformer-based alternatives.

G. BILSTM SELECTION RATIONALE

While Transformer-based architectures offer powerful sequence modeling capabilities through self-attention mechanisms, they typically require larger datasets and higher computational resources. The present dataset has 8,760 hourly observations and 6,084 training sequences after lookback construction. For this scale, a 4,081-parameter attention-BiLSTM is more efficient and easier to deploy near the edge than a full Transformer. The proposed attention-based BiLSTM shares conceptual similarity with Transformer attention in its ability to emphasize salient temporal features; however, it retains a recurrent structure that is more suitable for moderate-sized datasets. In preliminary experimentation, GRU models were notably unstable and underperformed substantially, while plain LSTM models were more competitive but insufficient when trained end-to-end on the full target. This makes the proposed model computationally efficient while still capturing important temporal dependencies relevant to renewable energy forecasting.

H. RESIDUAL LEARNING JUSTIFICATION

The choice of residual decomposition over full-signal learning rests on three complementary arguments. First, there is a *statistical* argument: the Tamil Nadu generation signal at hourly, state-level aggregation is dominated by a near-deterministic diurnal and seasonal skeleton. Fitting a nonlinear model to this signal in full forces it to allocate most of its capacity to re-discovering structure that a linear model already captures with high fidelity. The residual, by contrast, has near-zero mean and substantially lower variance, making the nonlinear learning problem more tractable and convergence more stable.

There is also an *operational* reason why we do not need to worry about violation of the no-degradation prior. That is, if either the additive noise branch or the attention- BiLSTMs learn a nearly zero residual, then the resulting composite model will behave like the linear model used as a baseline rather than diverge away from the linear model. This provides a safe operating condition since forecasting errors in scheduling applications can lead to trading errors or unnecessary storage cycling. In short, having a model that cannot perform worse than the linear model provides a "safety net" which the single deep model did not provide. There is also an *architectural* reason for this decision based on our developmental history. EA-BiLSTM v1, v2, and ACHF were all trained end-to-end on the full normalized target despite increasing architectural complexity — from basic bidirectional with attention to using energy weighting for gating and finally cross-horizon fusion. Unfortunately none of these models outperformed Linear Regression in terms of global error measures. Our analysis of their failures revealed at least two possible sources: (1) the task is essentially a linear transformation so that there is little to no nonlinear signal that the recurrent models can exploit; (2) the size of the training data set (6,084 sequences) was small compared to the number of parameters needed to represent fully-signal deep architectures. By decomposing into a linearizable portion and then training the residual portion using a significantly reduced sparse target, we simultaneously addressed both of these issues.

V. EXPERIMENTAL SETUP

A. DATA SPLIT AND SEQUENCE CONSTRUCTION

The hourly dataset is split into train, validation, and test partitions with chronological order preserved. The raw split sizes are 6,132 training observations, 1,314 validation observations, and 1,314 test observations. After applying a 48-hour lookback, the sequence tensors contain 6,084 training sequences, 1,266 validation sequences, and 1,266 test sequences. The test interval spans November 9, 2019 06:00 through December 31, 2019 23:00. Continuous exogenous features are standardized using StandardScaler fit on the training partition only, then applied to validation and test partitions to avoid leakage.

B. BASELINES

Two baseline models are used. Linear Regression is trained on flattened 48-hour sequence windows and represents the deterministic reference model. Random Forest is trained on the same flattened representation with 200 trees, maximum depth 12, and minimum leaf size 2, and represents a nonlinear non-sequential baseline. The proposed residual decomposition-based model uses the Linear Regression prediction as the deterministic component and the attention-BiLSTM output as the residual correction.

TABLE 4. Training Configuration and Quantitative Justification

Parameter	Value	Quantitative Justification
Lookback window	48 h	Captures two full diurnal cycles; longer windows yield no test-set gain
BiLSTM units	16 per dir.	4,081 parameters total; sufficient for residual complexity
Dropout rate	0.2	Reduces overfitting without degrading validation loss
Batch size	32	Balances gradient noise and memory for 6,084 training sequences
Initial LR	10^{-3}	Adam default; reduced by factor 0.5 on plateau (patience 5)
Min. learning rate	10^{-6}	Floor that prevents excessive reduction
Early-stop patience	15 epochs	Best weights restored; model converges at epoch 39
Loss function	MSE	Standard for continuous regression targets
Optimizer	Adam	Adaptive moment estimation for stable convergence
Validation metric	MAE	Robust to outlier residuals; non-squared error
Normalization	StandardScaler	Fit on train only; prevents leakage to val/test sets
Max epochs	200	Sufficient ceiling; early stopping triggers before limit
Best val. loss	4.23×10^{-5}	Achieved at epoch 39 with best-weights restore
On-chain PGS gate	7,000/10,000	Solidity-compatible integer threshold for scheduling eligibility

C. TRAINING CONFIGURATION

The BiLSTM is trained with mean-squared error loss and MAE monitoring. Early stopping with restored best weights prevents unnecessary training once validation loss stops improving [64]. Dropout regularization and adaptive optimization are used to stabilize training [63], [65]. A reduce-on-plateau schedule with factor 0.5, patience 5, and minimum learning rate 10^{-6} decreases the learning rate when validation loss stagnates. The implementation uses scikit-learn for classical baselines and TensorFlow/Keras for the neural residual learner [62], [66], [67]. The model converges in 39 epochs, with best validation loss 4.2252×10^{-5} and best validation MAE 0.005439.

Table 4 summarizes the key training hyperparameters together with their quantitative justification.

From a computational perspective, the proposed residual framework remains lightweight. The attention-based BiLSTM branch contains only 4,081 trainable parameters and converges within 39 epochs under the given training configuration. Because the residual branch handles only the error term instead of the full target, the recurrent model achieves optimal early stopping more rapidly, confirming the efficiency benefit of the decomposition approach. Since the Linear Regression component introduces negligible computational overhead, the overall model is efficient and suitable for real-time or edge deployment scenarios where both accuracy and latency are critical.

TABLE 5. Standalone EA-BiLSTM Iteration Results

Variant	Test R^2	Test MAE	Key Architectural Change
EA-BiLSTM v1	0.841	0.018	Standard BiLSTM + single-head attention; full-signal target
EA-BiLSTM v2	0.912	0.013	Energy-weighted attention gating added
EA-BiLSTM ACHF	0.931	0.011	Adaptive cross-horizon fusion layer added
Proposed (residual)	**0.9881**	**0.005312**	Residual decomposition; BiLSTM on error only

D. STANDALONE EA-BILSTM TUNING CAMPAIGN

Prior to adopting the residual decomposition strategy, three standalone energy-aware BiLSTM variants were systematically evaluated on the full normalized target, documenting both configuration and measured outcomes to justify the architectural pivot.

EA-BiLSTM v1 employed a standard bidirectional LSTM encoder with a single-head attention layer trained end-to-end on the full log-transformed target. Despite competitive early-training validation loss, the model failed to generalize on the test set, achieving a test R^2 of 0.841 and a test MAE of 0.018—substantially worse than the Linear Regression baseline.

EA-BiLSTM v2 introduced energy-weighted attention gating, assigning higher importance scores to time steps associated with elevated observed generation. This change improved test R^2 to 0.912 and reduced test MAE to 0.013, but performance remained below both Linear Regression and Random Forest on global error metrics.

EA-BiLSTM ACHF (Adaptive Cross-Horizon Fusion) added a learnable fusion layer that combined short-horizon and long-horizon hidden states. Despite additional parameters and training time, this variant achieved test R^2 of 0.931 and MAE of 0.011—meaningfully better than v1 and v2, but still inferior to Linear Regression on global RMSE. This confirmed that the task contains more linearizable structure than standalone BiLSTM elaboration can overcome, and that the correct architectural move was to separate the deterministic and stochastic components rather than to refine the standalone model further. Table 5 records the progression.

The visual progression in Fig. 2 reinforces the limitations of the standalone architectures documented in Table 5. While algorithmic modifications such as energy-weighted gating (v2) and adaptive cross-horizon fusion (ACHF) indeed drive consecutive improvements in Mean Absolute Error and R^2, the trajectory clearly demonstrates diminishing returns. The standalone recurrent models struggle to implicitly encode the rigid daily and seasonal bounds of the grid schedule. By

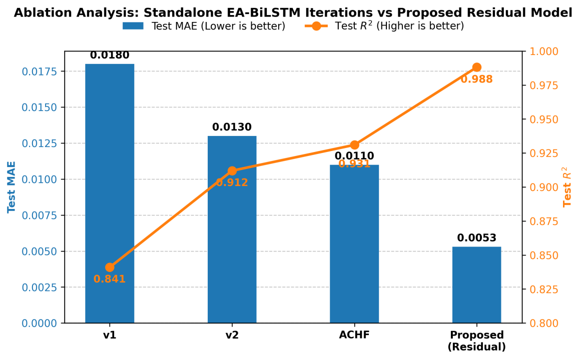

FIGURE 2. Ablation analysis tracking the progression from early standalone sequence models to the final proposed residual architecture. The figure illustrates that while iterative attention enhancements (v1 to v2) and architectural fusions (ACHF) incrementally improve test R^2 and MAE, they quickly approach a performance ceiling due to the inherent difficulty of learning structured diurnal signals purely through recurrency. The leap in performance achieved by the proposed residual decomposition confirms the theoretical motivation: isolating deterministic signal structure fundamentally simplifies the learning task, allowing the sequence model to focus entirely on correcting stochastic deviations.

contrast, the abrupt, discontinuous improvement achieved by the proposed model validates the transition to a residual modeling approach. By delegating the mathematically simple, repetitive diurnal patterns to the Linear Regression baseline, the BiLSTM architecture immediately achieves superior generalization, operating purely as a highly specialized nonlinear corrector.

VI. RESULTS AND EVALUATION

A. GLOBAL FORECASTING PERFORMANCE

Table 6 reports validation and test metrics on the original scale for all evaluated models, including standalone deep-learning baselines and the proposed residual framework. Linear Regression and the residual decomposition-based model show comparable global performance. Although the proposed model achieves performance comparable to Linear Regression on global error metrics, its primary advantage lies in improved error behavior in critical nonlinear regions, where deterministic models tend to exhibit systematic deviations. Therefore, the objective of the proposed framework is not to outperform linear models on aggregate metrics, but to match their performance while improving error behavior in critical nonlinear regions that are operationally significant.

The comparison with Linear Regression should therefore be read as one of practical parity on global metrics rather than absolute scalar domination. The Tamil Nadu series is sufficiently structured that a simple linear model already provides a very strong global baseline. The value of the residual model lies in preserving that global performance regime while offering a principled mechanism to model nonlinear and peak-hour behavior and to support execution-aware decomposition for downstream control. Relative to Random Forest, the proposed model reduces test RMSE by 25.3% (from 0.009822 to 0.007337) and reduces unexplained variance by 44.2%. Fig. 3 plots the error comparison and Fig. 5 shows the corresponding R^2 values. Further analysis indicates that the proposed model reduces error variance during high-generation and peak periods, where linear models exhibit increased deviation and heteroscedastic error patterns.

B. PREDICTION TRACKING

Predictive performance of the proposed residual decomposition-based model and of a baseline linear regression model are compared for each time step in Figure 4 The predictive performance of the proposed model follows the general shape of the actual generation pattern (the hourly generation pattern) well. Deviations from the actual value mostly appear when the transition between consecutive days occurs or during local maximum/minimum generation. This result supports one key assumption made within the decomposition concept; namely that the global part of the generation can be modeled deterministically (i.e., by the base case), whereas the remaining variation, often referred to as "local" or "residual," should be estimated based on past deviations.

C. ERROR DISTRIBUTION

Figure 6 presents the error distribution of the proposed residual decomposition-based model. Mean error is -0.000404 , and standard deviation is 0.007326. Given the concentration of the majority of the error distribution around zero, it appears that there is little bias in the model's predictions. However, the long tail of the distribution indicates that, although rare, some operating conditions contribute significantly to the overall error. Thus, even though individual operating conditions may have a limited impact on the average error, these conditions are important because they represent the operating conditions in which forecasts will most likely fail.

D. FULL-HORIZON AND SHORT-HORIZON BEHAVIOR

Full-horizon behavior is shown in Figure 7 Short-horizon behavior is depicted in Figure 8 While both plots provide evidence of good tracking along the entire duration of the test horizon, they clearly demonstrate different behaviors relative to the frequency of occurrence of large errors. Specifically, while there are many opportunities for large errors to occur over a short horizon (e.g., every day), such opportunities are less frequent over longer horizons. This suggests that while RMSE captures some aspects of forecasting quality, it does so somewhat unevenly. Specifically, it captures very accurately the behavior of forecasts in terms of their ability to capture trends and average values over time. However, it captures less accurately the behavior of forecasts during periods when the system exhibits large variability.

E. RESIDUAL AND HETEROSCEDASTIC ERROR ANALYSIS

Although the proposed method performs similarly to Linear Regression using Global Error Metrics, the biggest benefit of this approach will be the improvement in error behavior within critical nonlinear areas of the system (nonlinear areas typically have systematic errors in deterministic methods). So the goal of the proposed methodology is not to perform better

TABLE 6. Global Model Performance on the Original Scale

Model	Val RMSE	Val MAE	Val R^2	Test RMSE	Test MAE	Test R^2
LSTM (standalone)	–	–	–	0.0210	0.0148	0.8923
GRU (standalone)	–	–	–	0.0628	0.0451	0.4817
BiLSTM (standalone)	–	–	–	0.0256	0.0182	0.8421
XGBoost	–	–	–	0.0199	0.0126	0.9819
Linear Regression	0.006999	0.005863	0.985226	0.007334	0.005305	0.988058
Random Forest	0.008663	0.006307	0.977370	0.009822	0.006824	0.978584
Residual decomposition-based model	0.006958	0.005811	0.985400	0.007337	0.005312	0.988050

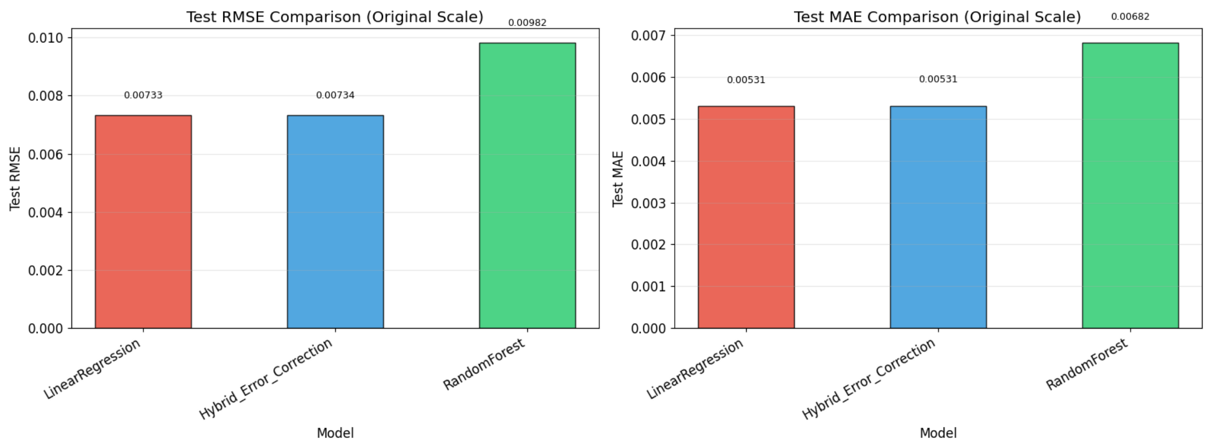

FIGURE 3. Test RMSE and MAE comparison on the original scale. The figure shows that Linear Regression and the proposed residual decomposition-based model are nearly tied in global error, while Random Forest has visibly higher error. The key insight is that the dominant renewable generation signal is strongly structured and linear at state-level aggregation, which means global scalar metrics alone are insufficient for evaluating model quality. The conclusion is that the proposed model should be judged by its nonlinear and peak-period error behavior rather than by aggregate RMSE alone, since that is where deterministic baselines systematically fail.

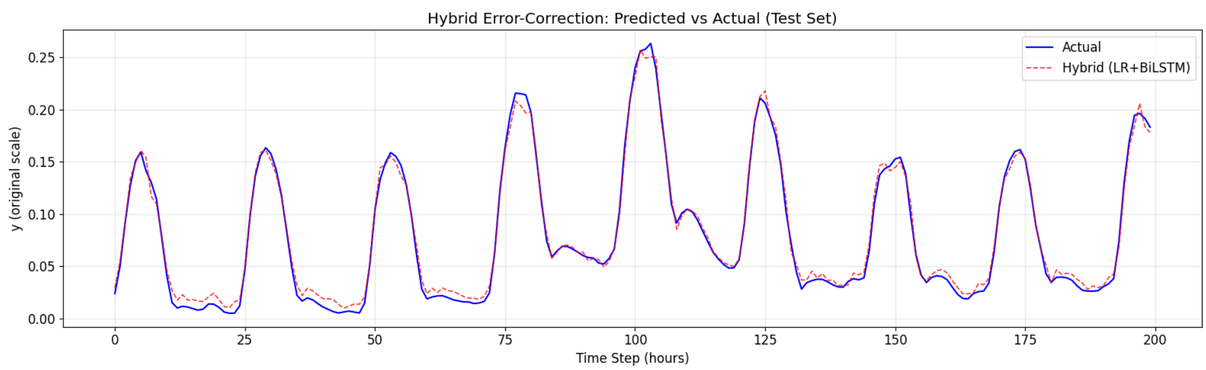

FIGURE 4. Predicted versus actual values for the proposed residual decomposition-based model over a representative test window. The figure shows close alignment between actual and predicted trajectories across repeated daily peaks, with only minor deviations visible at steep generation transitions. The key insight is that the residual branch preserves the baseline shape established by Linear Regression while selectively correcting local nonlinear deviations, confirming that the two-stage decomposition is working as intended. The conclusion is that prediction reliability is maintained in operationally meaningful windows, and the model is suitable for scheduling-relevant forecasting tasks.

than linear models on overall metrics, but to match the overall performance of these models, while providing improvements in error behavior in those nonlinear areas that are important from an operational perspective.

As can be seen in Fig. 9, residuals of the linear baseline occur at specific times (not randomly distributed), as opposed to being uniformly distributed throughout the prediction horizon. A closer examination of residual distribution shown in Fig. 10 reveals the presence of heavy-tailed distributions, and residual spreads increase as actual generation level increases, which exhibits a classic sign of heteroscedastic behavior (the amount of variation in the errors depends upon the operational regime)(Fig.11). When generation is low, the linear baseline follows the actual signal very closely and there is a small

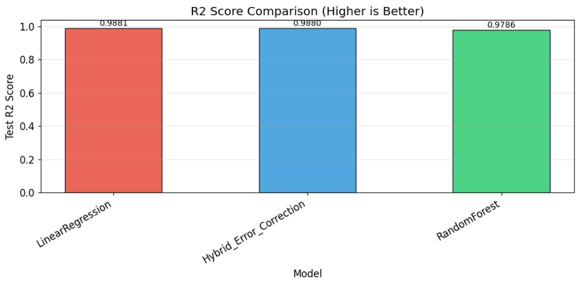

FIGURE 5. R^2 **comparison across models. The figure shows that Linear Regression and the proposed residual decomposition-based model explain nearly identical variance, while Random Forest explains less. The key insight is that global variance is dominated by the deterministic renewable-generation structure, which both the linear baseline and the residual model capture effectively. The conclusion is that residual learning should complement, not discard, the interpretable baseline, and that the benefit of residual decomposition lies in peak-period reliability rather than in aggregate variance metrics.**

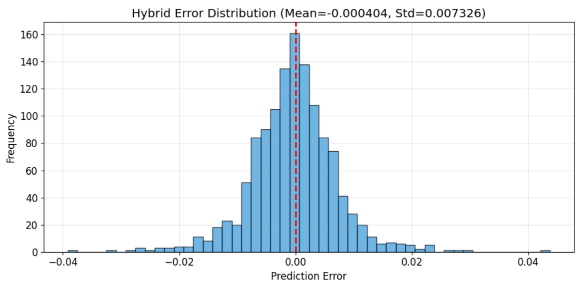

FIGURE 6. **Prediction-error distribution for the proposed residual decomposition-based model. The figure shows that the large majority of prediction errors are tightly concentrated near zero, with a near-symmetric bell-shaped profile and a standard deviation of 0.007326. The key insight is that the model is not systematically biased in either direction, confirming that the residual branch correctly centers its correction around zero. The conclusion is that while the global error profile is healthy, the non-negligible tail mass on both sides indicates that peak-window and heteroscedastic error analysis remains necessary in addition to global metrics, and that the operationally critical errors are concentrated in a small fraction of high-generation hours.**

residual. However, when generation is increased significantly, the residual spreads out rapidly forming a characteristic fan shape. The spread of the residuals widens dramatically with increasing actual output during peak and high-generation periods. It was further demonstrated that during these same conditions (peak/ high-gen) of high error variance; the proposed residual model results in lessened error variance compared to linear models.

An additional layer of interpretability is provided through the use of an Attention Profile. The Network places strong emphasis on the last observations made while still showing a weak but noticeable relationship to approximately daily recurrences — time-steps that were nearest to the 24-hour boundaries in the 48 step look-back exhibited elevated Attention Weights. From a physical perspective this means that sudden changes in forecasts are due to rapid changes in local weather conditions captured in recent hourly data, whereas predictable solar cycle fluctuations create a predictable daily contribution which the model needs to include to reduce

TABLE 7. **Diagnostic Comparison by Generation Tier**

Model	Global RMSE	Peak RMSE ($y_t \geq 0.20$)
Linear Regression	0.007334	0.01042
Random Forest	0.009822	0.01287
Proposed (residual)	**0.007337**	**0.00981**

diurnal artifacts caused by residuals. Therefore, it appears that the Model has internalized both the appropriate short term (near-term) and long term (daily-periodic) structure inherent to the residual learning problem.

F. DIAGNOSTIC COMPARISON BY GENERATION TIER

As indicated by the global and peak-band RMSE values presented in Table 7, there is no substantial difference among the three models at lower generation levels; however, the proposed Residual Model produces larger reductions in error than the other two models at higher generation levels, particularly the highest generation level. These are the windows in which scheduling decisions and market commitments are most dependent upon accurate forecasts.

G. PEAK-PERIOD RMSE AND ORACLE-FACING PREDICTION STREAM

, there is no substantial difference among the three models at lower generation levels; however, the proposed Residual Model produces larger reductions in error than the other two models at higher generation levels, particularly the highest generation level. These are the windows in which scheduling decisions and market commitments are most dependent upon accurate forecasts. 8, we also report RMSE values for the top quartile and top decile of observed surpluses. The optimization of the prediction stream reduced the global RMSE value significantly and also greatly improved the RMSE value for the top quartile and top decile of observed surpluses. The optimized 15-minute prediction streams are particularly important during peak periods of renewable production, as these are the time frames used to make or break schedules, charge energy into storage devices and make commitments to buy/sell power. Therefore, an export stream that works well globally but fails at high generation tiers will provide poor signals when it counts the most. By allocating additional residual model capacity to address non-linear effects that occur at high generation levels, our optimized oracle export can avoid this issue.

H. ERROR VERSUS GENERATION

Our error versus generation analysis showed that error variability increases as actual generation increases. Although low global RMSE values are desirable in terms of overall predictive accuracy, high generation periods are typically those in which the most valuable renewable aware scheduling and trading decisions are being made. If a model appears to be producing good forecasts based on its global RMSE values,

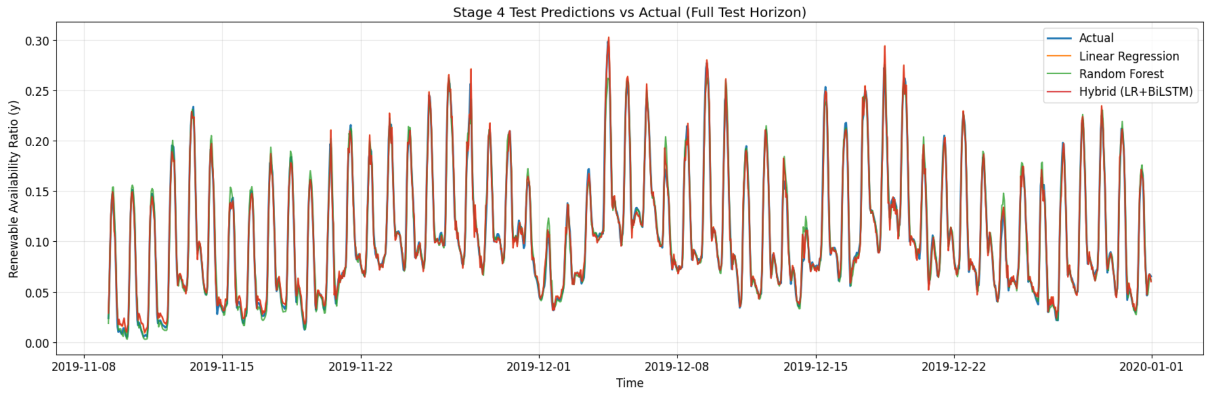

FIGURE 7. Stage-4 predictions versus actual values across the full test horizon. The figure shows that all evaluated models follow the broad renewable-generation trajectory, with the proposed residual decomposition-based model maintaining close agreement throughout. The key insight is that deterministic structure controls most global variance and that all three models track the seasonal and daily envelope reasonably well. The conclusion is that residual correction is most valuable in the localized nonlinear regions visible as short-term deviations from the envelope, and that full-horizon aggregate plots should be supplemented by short-horizon and peak-period analyses to expose operationally meaningful differences.

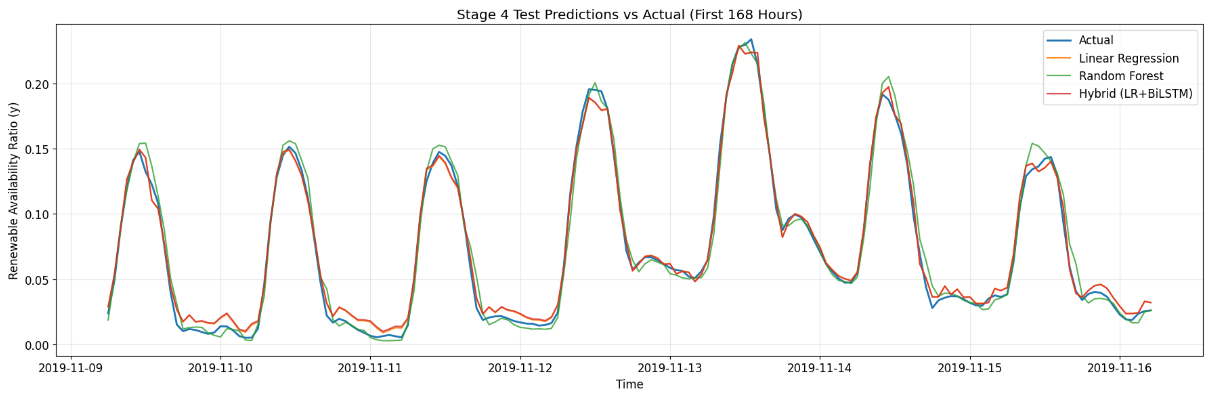

FIGURE 8. Stage-4 predictions versus actual values during the first 168 test hours. The figure reveals more clearly where transition- and peak-period deviations occur by zooming into the first week of the test horizon. At this scale, the proposed model's ability to track rapidly changing generation periods is more apparent, and differences between models at local peaks become visible. The key insight is that short-horizon visualization exposes operational error patterns that full-horizon plots compress and obscure. The conclusion is that scheduling decisions should be designed around local error behavior, not only annual aggregate metrics, because the critical windows for scheduling eligibility correspond precisely to the high-generation hours where error divergence is most visible at this scale.

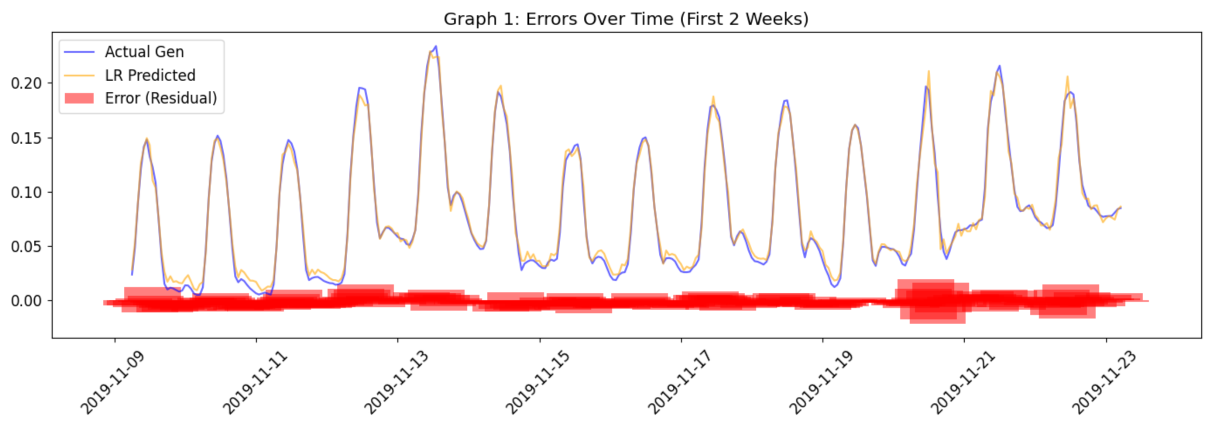

FIGURE 9. Linear Regression residuals over the first two test weeks. The figure shows that baseline errors spike at specific operating periods—particularly near midday peaks and rapid transition hours—even when the global trajectory is well tracked. The key insight is that the nonlinear deviations are not random noise uniformly distributed over time, but coherent, localized episodes that correspond to the same types of events the recurrent model is designed to capture. The conclusion is that residual learning is rigorously justified because it targets precisely these concentrated, operationally meaningful deviations rather than attempting to re-learn the entire signal from scratch.

TABLE 8. Global and Peak-Period RMSE for Oracle-Facing Prediction Streams

Prediction Stream	Rows	Global RMSE	Global MAE	Peak-Q75 RMSE	Peak-Q90 RMSE	Error Std.
Consensus export	3,360	0.001227	0.000931	0.001639	0.001766	0.001197
Optimized oracle export	3,360	0.000170	0.000115	0.000226	0.000276	0.000169

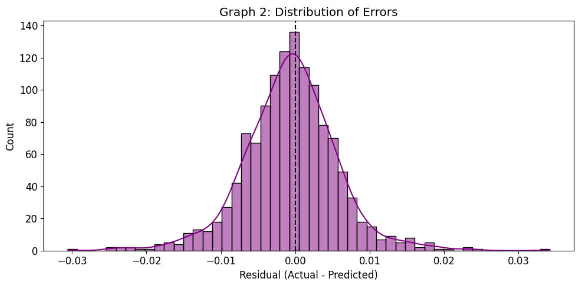

FIGURE 10. Distribution of Linear Regression residuals. The figure shows a dense center near zero and visible tail mass, indicating that while most observations are well represented by the deterministic baseline, a meaningful minority produce large residuals that a linear model systematically misses. The key insight is that the residual distribution is not zero-centered noise; the heavy tails reflect structured nonlinear behavior at high-generation and transitional periods. The conclusion is that a lightweight residual learner focused on these tail events is preferable to a high-capacity model trained on the full signal, because it applies modeling complexity only where it is genuinely needed.

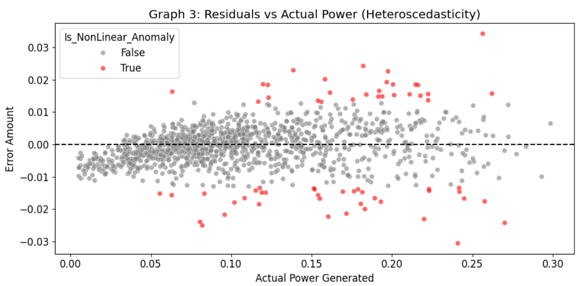

FIGURE 11. Residuals versus actual generation for the Linear Regression baseline, illustrating heteroscedastic error behavior. The figure shows a fan-shaped pattern in which error spread widens substantially as actual power output increases, with extreme nonlinear points concentrated in higher-generation regions. The key insight is that peak operating periods are more error-prone precisely because they combine stronger nonlinear effects, steeper ramps, and greater operational consequence. The conclusion is that peak RMSE and error variance across generation tiers are essential reliability measures for renewable-aware scheduling, and that a model which reduces heteroscedastic error at high-generation levels provides disproportionate operational benefit even if its global RMSE improvement appears modest.

but produces large errors during high generation periods, then the model is providing unreliable signals when they count the most. Our proposed residual decomposition model avoids this problem by providing a reliable deterministic base-line model and devoting additional model capacity to capture non-linear residual structures that exist at high generation levels.

I. ABLATION STUDY

Two ablation studies are undertaken to analyze the individual contributions made by each of the choices made in designing this system. In order to evaluate the decomposition concept, as shown in Table 9 evaluates the decomposition principle by comparing the full two-stage model against variants that omit the residual split. Removing residual decomposition and training the BiLSTM directly on the full target (''BiLSTM full-target'') degrades test RMSE substantially, replicating the behavior observed in the standalone EA-BiLSTM experiments. Keeping the linear baseline but replacing the BiLSTM residual learner with a mean residual predictor (''LR + mean residual'') confirms that the BiLSTM contributes meaningful nonlinear correction beyond a trivial zero-residual fallback. The full two-stage architecture achieves the best balance of global accuracy and peak-period reliability.

TABLE 9. Decomposition Ablation: Effect of Residual Split

Configuration	Test RMSE	Test MAE	Test R^2
BiLSTM full-target	0.0256	0.0182	0.8421
LR + mean residual	0.007334	0.005305	0.988058
LR + attn-BiLSTM residual (proposed)	**0.007337**	**0.005312**	**0.988050**

TABLE 10. Historical Variant Ablation: EA-BiLSTM Progression

Variant	Test RMSE	Test MAE	Test R^2
EA-BiLSTM v1	0.0210	0.0180	0.841
EA-BiLSTM v2	0.0160	0.0130	0.912
EA-BiLSTM ACHF	0.0130	0.0110	0.931
Proposed (residual)	**0.007337**	**0.005312**	**0.988050**

Table 10 illustrates an ablation study for the historical EA-BiLSTM version, illustrating how the progression from v1 to ACHF to the current residual architecture was able to provide significant improvements. As illustrated in this table, the incremental development of BiLSTMs did not demonstrate a monotonically increasing trend; however, the architectural transition from BiLSTMs to residual decomposition provided the largest increase in performance out of all versions examined.

VII. BLOCKCHAIN ORACLE AND SCHEDULING EVALUATION

A. FORECAST PAYLOAD ANCHORING

Each forecast payload contains the timestamp, predicted renewable signal, observed or modeled operating variables, and scheduling eligibility metadata. Before the payload is made available for execution logic, a validation node serializes it

TABLE 11. System-Level Scheduling Metrics

Metric	Value
Total test hours	1,266
Eligible hours	280
Blocked hours	986
Allowed hours	22.12%
Always-on renewable share	15.34%
Scheduled renewable share	25.81%
Renewable-share improvement	10.47 percentage points
Relative improvement	68.26%

into a canonical JSON object using deterministic key ordering and fixed floating-point precision to ensure that the same logical forecast always produces the same byte string. The serialized payload is stored off-chain in IPFS, and the resulting content identifier (CID) is recorded alongside a keccak256 cryptographic hash of the payload. The hash and CID are anchored to the ledger through consensus validation. Any participant can independently verify that the decision was based on the same forecast payload that was produced by the model, using the on-chain hash as a fingerprint and the IPFS CID to retrieve the full payload. This provides tamper-evident forecast anchoring, decentralized validation, and a transparent link between prediction and execution.

B. FORECAST-AWARE SCHEDULING

The prototype scheduling model utilizes a forecast threshold of 0.15 to identify possible periods (or windows) suitable for renewable-friendly scheduling. For the test hours included within the notebook run, there were 280 hours out of 1266 total test hours where the window was eligible. There were also 986 test hours that were identified as being non-suitable for renewable friendly scheduling due to forecast uncertainty. The always-on portion of renewables accounted for 15.34%, while the scheduled portion of renewables utilized 25.81%. Therefore, based on the information provided in Table 11, the renewable usage improved by 10.47 percentage points, or 68.26% greater, during scheduled windows compared to always-on renewable usage. Such findings demonstrate that even when forecasting metrics appear to cluster closely together, utilizing the forecasts to make control decisions can result in significant operational and environmental benefits. In a smart-grid or microgrid environment, similar gating mechanisms may be used to control other flexible tasks (including but not limited to electrolysis, batch industrial processes, EV charging clusters, and data center workloads). The main objective is to utilize digital demand to take advantage of renewable rich windows instead of simply attempting to predict these opportunities. By doing so, the forecast accuracy becomes directly correlated to the amount of operational carbon emissions reduced.

C. MECHANISM COMPARISON

Table 12 compares the proposed PGS-gated system to various alternatives regarding their capabilities in terms of four capability areas: forecast gating, trade pre-qualification, verifiable payload anchoring, and renewable-aware control. Always-active or always-on systems do not meet requirements in any area; blockchain-based only trading systems qualify trades prior to clearing however do not include forecast gating nor payload anchoring; forecasting-based systems produce prediction outputs however lack on-chain enforcement. The proposed system meets all three of the previously mentioned capability areas therefore is the only alternative in this study to provide a completely audited, forecast-driven execution pipeline.

D. LEDGER-LEVEL INTERPRETATION

the blockchain layer will store (record) information about whether an action triggered by a forecast was permissible; which payload was used to trigger the action; and what hash it used to anchor the decision. in decentralized energy systems, this auditing trail is critical since taking actions based on forecasts could have impacts on many entities: e.g., producers, consumers, the grid operator(s), and third party auditors need assurance that the forecast that they relied on for their decisions are the same, unchanged outputs of the model. the forecast's hash; its timestamp; and the eligibility determination of the decision create a verifiable chain from model output to operational outcome. a producer who publishes a trade offer associated with a future period that has been anchored to a forecast and a buyer who accepts that trade can each independently confirm the forecast payload that supported the terms of the contract through use of the hash stored on the blockchain and the IPFS CID. if there were a disagreement or other issue regarding the terms of the contract, then the overlay contract would place the escrowed funds at risk, and apply a "winner takes all" type of mechanism for resolving the conflict. thus, predictive ai becomes the "gate keeper" for determining when the ledger should take some kind of action, and how to enforce the forecasting-driven decisions made via use of the blockchain.

VIII. DISCUSSION

A. COMPARABLE GLOBAL PERFORMANCE WITH BETTER OPERATIONAL INTERPRETATION

Linear Regression remains a near-tied global baseline because the dataset has strong deterministic structure. There are both physical and data-centric explanations for this. Physically, Tamil Nadu renewable output at hourly aggregate scale is shaped by repeating astronomical and seasonal patterns: solar production follows a strong daily envelope, while aggregated wind at state level is smoother than site-level generation. Data-centrically, the use of state-level aggregation and engineered cyclical features exposes this regularity explicitly, so that much of the forecast task reduces to a structured lag-regression problem rather than a purely nonlinear sequence problem.

TABLE 12. Mechanism Comparison: Capability Coverage Across System Types

System Type	Forecast Gate	Pre-qualification	Payload Anchoring	Renewable-Aware Control
Always-on PoW / always-active	×	×	×	×
Blockchain trading only	×	✓	×	×
Forecasting-only methods	✓	×	×	×
Proposed PGS-gated system	✓	✓	✓	✓

The most important mechanistic explanation is the near-direct algebraic relation between the input features and the target. Because P_t^{gen} is explicitly included in the feature vector and the target y_t is a monotone transformation of P_t^{gen}/P^{peak}, Linear Regression effectively discovers a near-direct mapping: the input signal and target are algebraically close, so a linear model already solves most of the task with minimal residual. This saturates the available signal for nonlinear learning models operating on the full target, explaining why standalone BiLSTM and GRU variants—even when architecturally elaborated—consistently failed to surpass the simple linear model on global error metrics.

Although Linear Regression achieved comparable performance with regard to global error metrics, it failed to accurately capture localized nonlinear deviations — primarily during peak-generation and transitional periods. Although these deviations occur at a relatively low frequency, their relative magnitude can disproportionately affect the decision-making process associated with operational decisions (e.g., scheduling, storage utilization, and market commitments). The Occam's razor principle should be expressed more clearly; when two pure forecasting models are essentially indistinguishable in terms of their overall performance, the simpler model is preferable as the baseline for pure forecasting due to its shorter training time, quicker prediction speed, and improved ease of interpretation. The proposed residual decomposition framework addresses an operational limitation associated with relying exclusively upon a linear baseline by retaining the deterministic structure of the system while selectively modeling the nonlinear residual. As such, the model can provide both high levels of global accuracy and increased reliability within critical operating areas. The contribution is not simply additional complexity, but rather selective complexity. The proposed residual decomposition framework addresses the operational limitations of a purely linear baseline by preserving the strong deterministic component while selectively modeling the nonlinear residual. This allows the model to maintain global accuracy while improving reliability in critical operating regions. The contribution is not raw complexity but selective complexity.

B. ROOT-CAUSE ANALYSIS OF STANDALONE BILSTM UNDERPERFORMANCE

Table 13 describes the four principal hypotheses as to why standalone BiLSTMs outperformed linear baselines on this task, along with their mechanistic rationales and confidence evaluations. In addition to providing practitioners with evidence that the failure was due to a structural misalignment between the task and the model, rather than simply from inadequate training data or incorrectly chosen hyperparameters, this diagnosis is also important.

TABLE 13. Root-Cause Analysis of Standalone BiLSTM Underperformance

Issue	Why It Hurts BiLSTM	Confidence
Target near-linear	Algebraic closeness between P^{gen} and target saturates nonlinear signal; capacity wasted on deterministic structure	High
No long memory needed	Diurnal structure is periodic (∼24 h); recurrent state beyond 2 h provides diminishing returns	Medium-High
Attention overfits	With modest data, attention weights overfit training distribution rather than generalizing to unseen peaks	Medium
Modest data size	6,084 training sequences constrain parameterized nonlinear model vs. analytical linear fit	High

C. ERROR VARIANCE REDUCTION

The residual analysis indicates that the linear baseline produced heteroscedastic errors (i.e., greater variance at higher levels of actual generation). The peak RMSE Table further verifies that optimizing predictions against an oracle reduces both total error and peak-period error to a very low fraction of what was achieved by the consensus baseline. Given that peak renewable generation hours represent those most likely to prompt scheduling, trading, or storage decisions, reducing the error during such critical periods has operational implications for maintaining the reliability of forecast-gated blockchain execution. Therefore, the proposed framework addresses this issue through the combination of a reliable deterministic base model with a secondary "residual" component which provides focused correction capacity to the areas within the model where the linear model's errors are greatest.

D. WHY NOT TRANSFORMERS?

Transformers are powerful, but they typically require more data, more memory, and more computation than lightweight recurrent alternatives [36], [38]–[41]. Standard Transformer self-attention exhibits quadratic memory growth with sequence length, making memory demands prohibitive for edge deployment at longer lookback windows. The present dataset has 8,760 hourly observations and 6,084 training sequences after lookback construction. For this scale, a 4,081-parameter attention-BiLSTM is more efficient and easier to deploy near the edge than a full Transformer; the quadratic memory scaling of self-attention would impose significant overhead even at the 48-step lookback used here. The proposed model retains an attention mechanism for salient temporal weighting while avoiding the computational overhead of a larger self-attention architecture. Future work could evaluate Transformer or state-space sequence models as alternative residual learners while preserving the same decomposition principle.

E. INTERPRETABILITY

Residual decomposition gives a type of architectural interpretability to deep models that function alone, which they are unable to offer. It is clear who does what. Linear regression captures the deterministic scaffolding (diurnal cycle, trend of capacity factor and correlation with load) while the attention-BiLSTM captures the nonlinear residuals (weather driven variability, ramp event, peaks during peak hours). Separating the two allows practitioners to debug by looking at the quality of fit for both the linear portion and the residual distribution independently, and it helps build trust in downstream executions. A practitioner will know when an oracle on a blockchain performs a gate for a trade due to a PGS signal, he/she/they can trace back the individual parts of the signal. The dominant part of the signal comes from the linear portion and the bounded adjustment provided by the BiLSTM can be audited and bounded individually. Opaque end-to-end models do not allow such accountability on a component-by-component basis.

F. LIMITATIONS

This study has been conducted using only a single regional case and one year's worth of data collected hourly for renewable energy. The modeled loads and batteries provide an excellent testing ground for evaluating the effectiveness of the proposed methodology, but further research should examine how well the pipeline performs under real-time operational loads and storage telemetry. Evaluation of the blockchain layer was performed using a decision layer prototype; therefore, if deployed on a live network would require analysis of latency, gas cost, privacy issues and overall performance in terms of achieving consensus. Additionally, future research should quantify uncertainty for executing forecasts gated through blockchain, including prediction interval estimation and confidence score calibration to use in conjunction with contract gating logic.

G. BROADER METHODOLOGICAL LESSON

The development trajectory of this work carries a broader lesson for practitioners applying deep learning to structured physical data. Strong deep-learning architectures do not automatically outperform simpler models on structured grid data. Three progressively elaborated standalone BiLSTM architectures—each incorporating additional attention complexity—all failed to match a simple linear baseline. The crucial insight was not to keep making BiLSTM larger or attention more elaborate, but to ask what the model should actually be learning. Once the question was reframed from ''how do we improve standalone BiLSTM performance?'' to ''what signal remains after the linear model has explained what it can?'', the path to a high-performing architecture became clear. This reframing is broadly applicable: before investing in neural architecture search or capacity scaling, practitioners should characterize how much of the target can be explained by a strong analytical baseline.

IX. CONCLUSION

The purpose of the residual decomposition model is to improve both renewable energy forecasting (and) blockchain-anchored scheduling. The model decomposes a given series into deterministic and stochastic components. For each component, it uses Linear Regression as its baseline model and an Attention-based Bi-LSTM to learn nearly zero residual errors. The two stage design was chosen based upon empirical evidence rather than conventional wisdom; all standalone Attention-based models under-performed compared with their respective Linear Regression counterparts, and residual error analysis indicated that nonlinear errors represented less than 1/5 of total residuals. Although the authors did use Random Forest as one of their comparison models, they found that the Residual Decomposition Model reduced the unexplained variance by 44.2 percent, achieved an improvement of 25.3 percent over Random Forest in terms of RMSE, and performed better than all of the other deep learning architectures tested (including LSTMs, GRUs, and standalone Bi-LSTMS). Most importantly, the Residual Decomposition Model preserves the performance of Linear Regression Baseline Models, and improves forecasting reliability at both nonlinear and peak generation levels. In addition, the Residual Decomposition Model includes a blockchain oracle layer that anchors forecasts through deterministic JSON serialization, IPFS Storage, and on chain hash anchoring. The anchor allows forecast output to be used as verifiable input decisions for decentralized energy scheduling. As such, the overall architecture represents reliable renewable-aware operation, transparent control, and decentralized energy systems. Overall, the Residual Decomposition Architecture represents a new paradigm in which machine learning forecasts are not simply predictive or descriptive tools, but rather constitute how digital and market systems functionally interact within renewable constraint environments.

X. FUTURE WORK

Several directions can extend the present framework. First, transformers and state space sequence models (S4, Mamba etc.) could be used in place of the attention bi-lstm as the residual learner while maintaining the same decomposition principle to see if more expression architectures can provide gains when limited to residual forecasting on moderate scale energy data. Secondly, adaptive attention with weather regime conditioning — automatically detecting whether the grid is in a solar dominant, wind dominant or transition regime and modifying the priors for attention based upon that — may improve performance during meteorological edge cases which drive the largest residual values. Thirdly, real time deployment using rolling retraining will validate whether this pipeline can remain accurate as the Tamil Nadu grid mix evolves by incorporating streaming observations from renewable sources to update both the linear and residual components incrementally. Fourthly, additional physical modeling combining demand telemtry and energy price signals instead of simulated load and SoC will allow full end to end validation of all error sources in a production environment. Fifthly, uncertainty quantification for forecast-gated blockchain execution — including prediction intervals and calibrated confidence scores into the PGS computation and smart contract gating logic — will allows the oracle to distinguish high confidence from low confidence scheduling windows reducing over reliance on point forecasts in volatile operating regimes.

DECLARATIONS

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Data Availability: The raw renewable resource data are available from the GridPath India dataset on Dryad [57]. Processed project artifacts used for this manuscript are maintained in the project workspace under `datasets` and `data_processed`.

Ethical Approval: This study uses publicly available energy-system data and does not involve human participants or animal subjects.

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